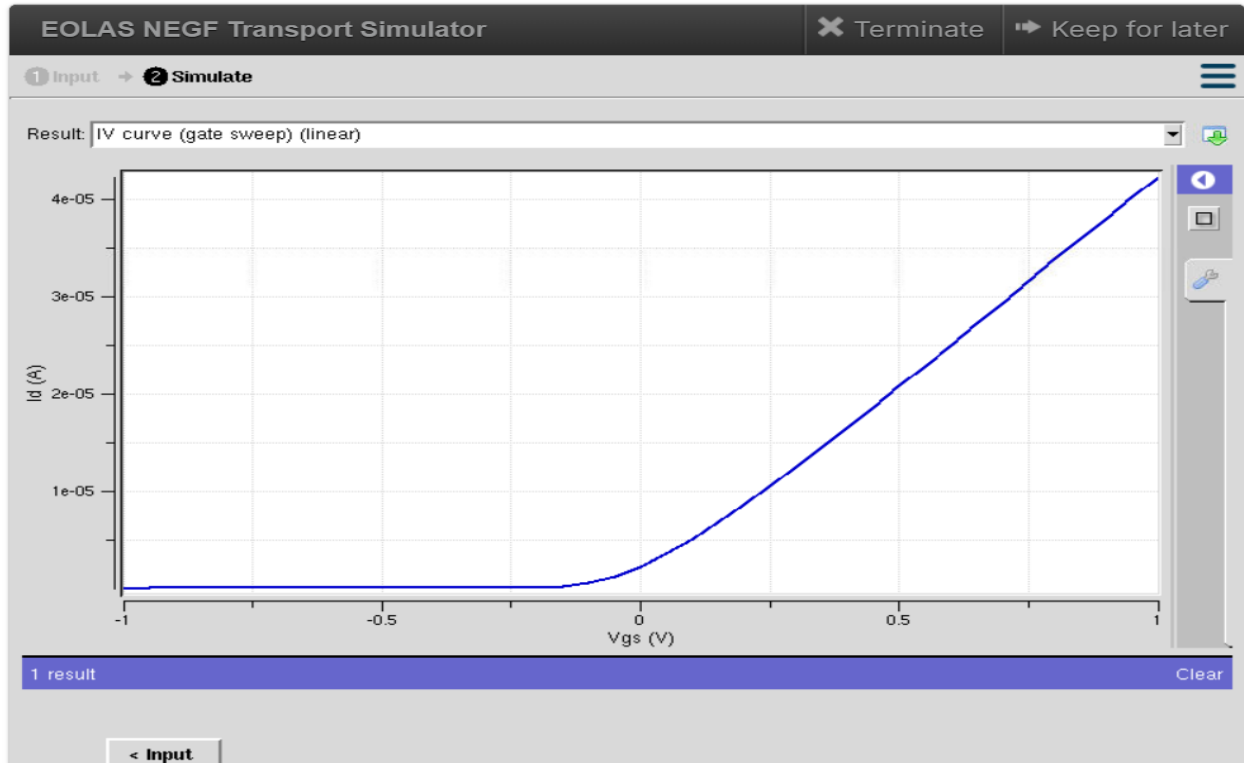
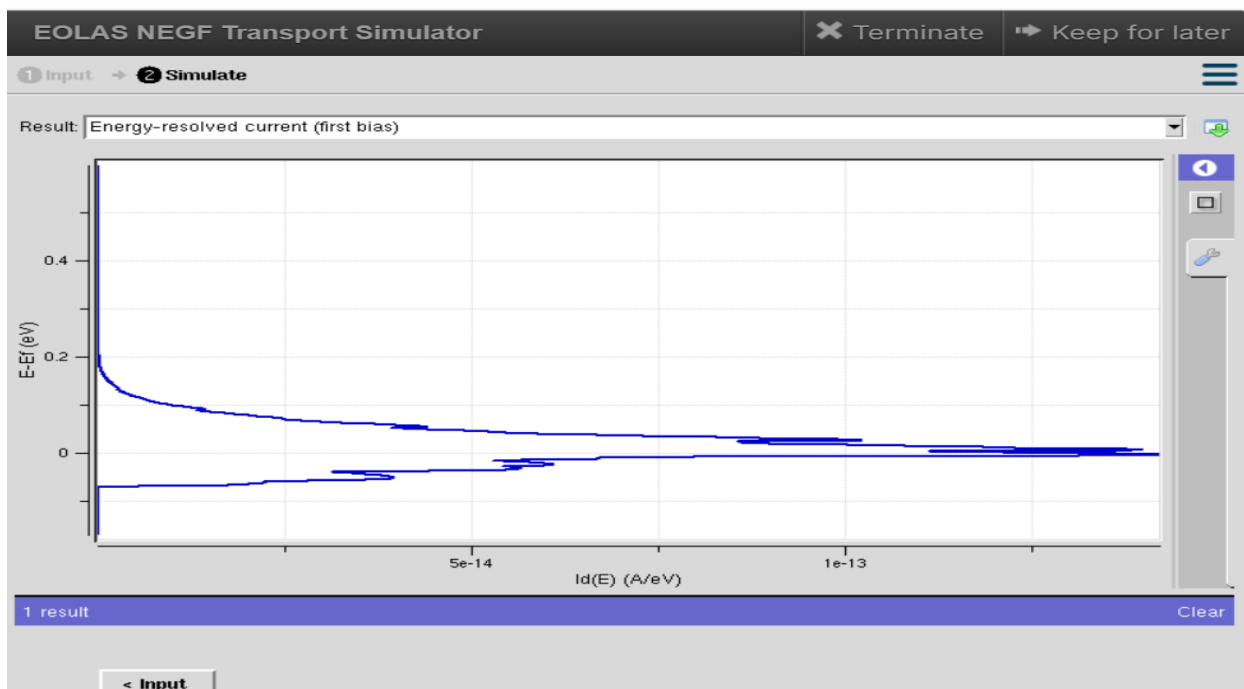




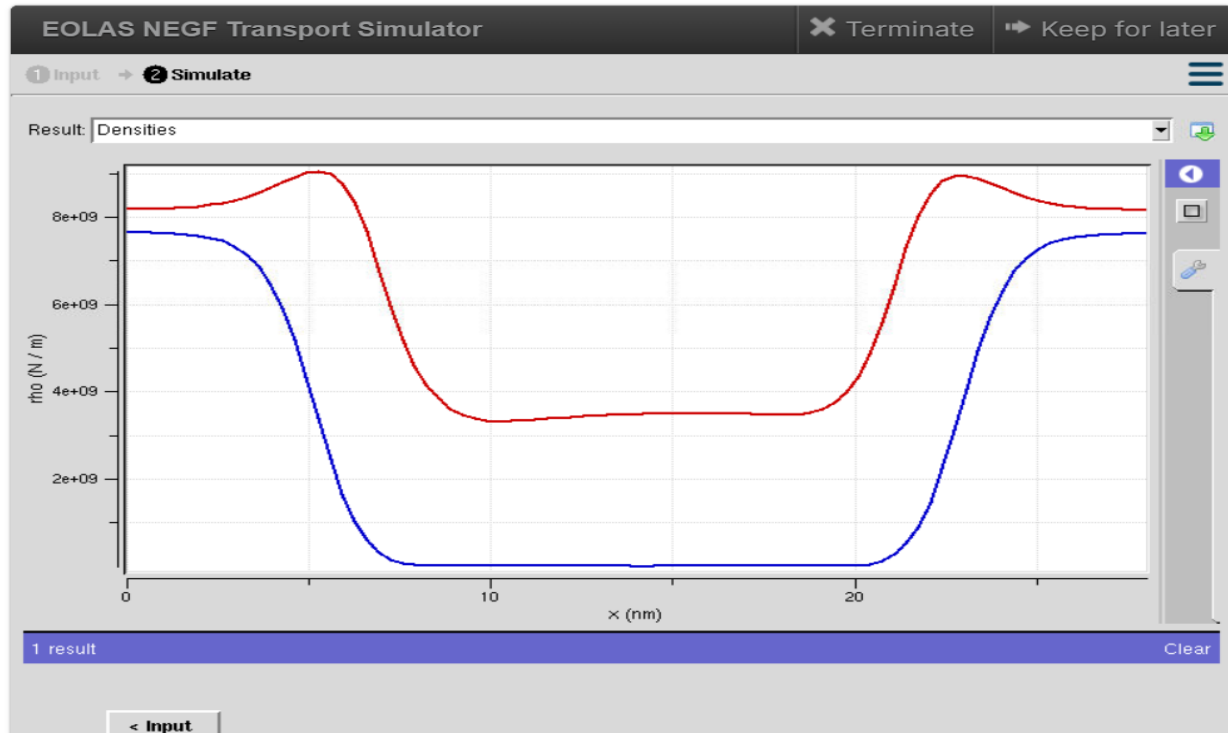
Finfet output is the Transfer Characteristics (I_d - V_{gs}) of a transistor. It shows how the drain current (I_d) changes with the gate-to-source voltage (V_{gs}) while keeping the drain voltage (V_{ds}) constant.

EXP 8 TEST CASE 1:



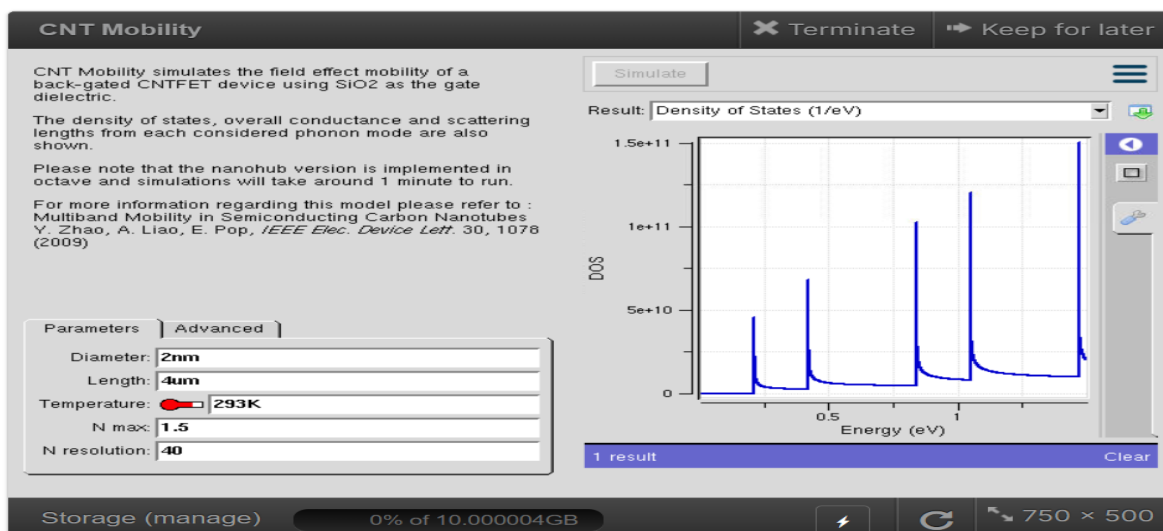
Graph shows how much current is carried by electrons at different energy levels under the applied bias voltage.

TEST CASE 2:

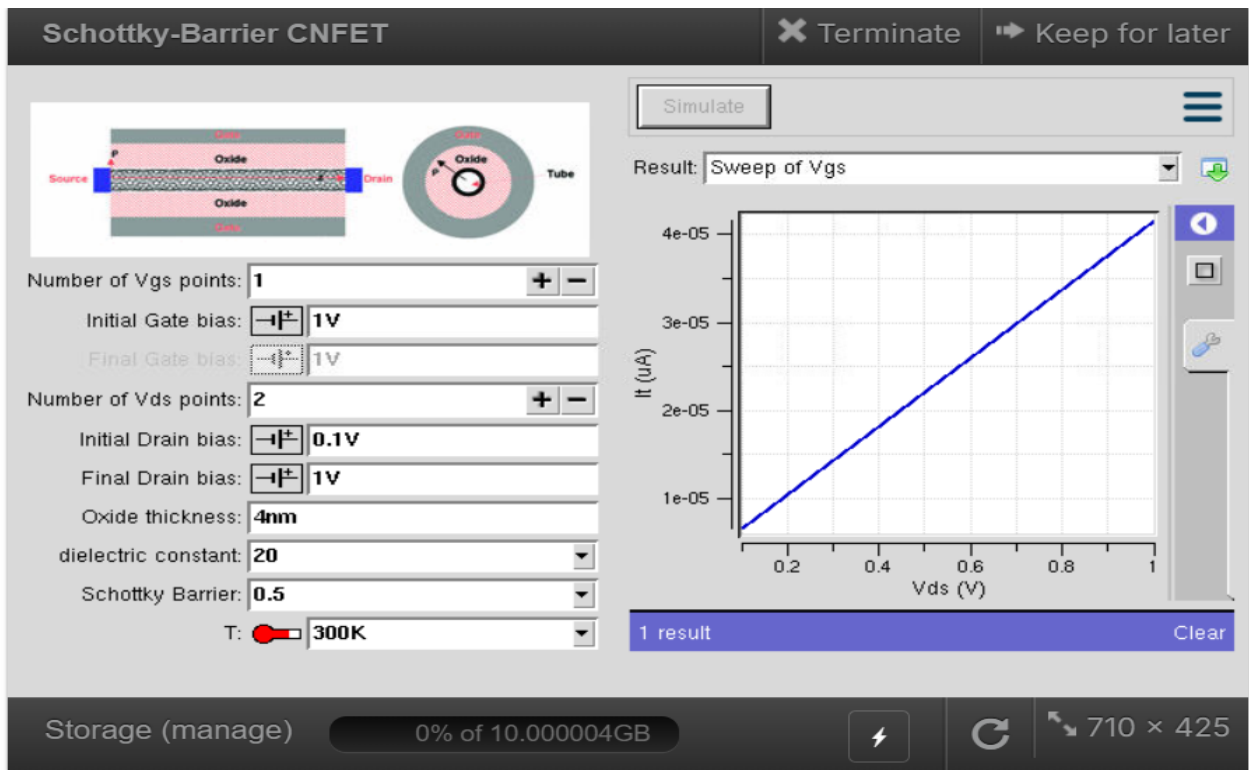


- **Red Curve:** Electron density
- **Blue Curve:** Hole density (or another carrier density distribution depending on the simulator settings)

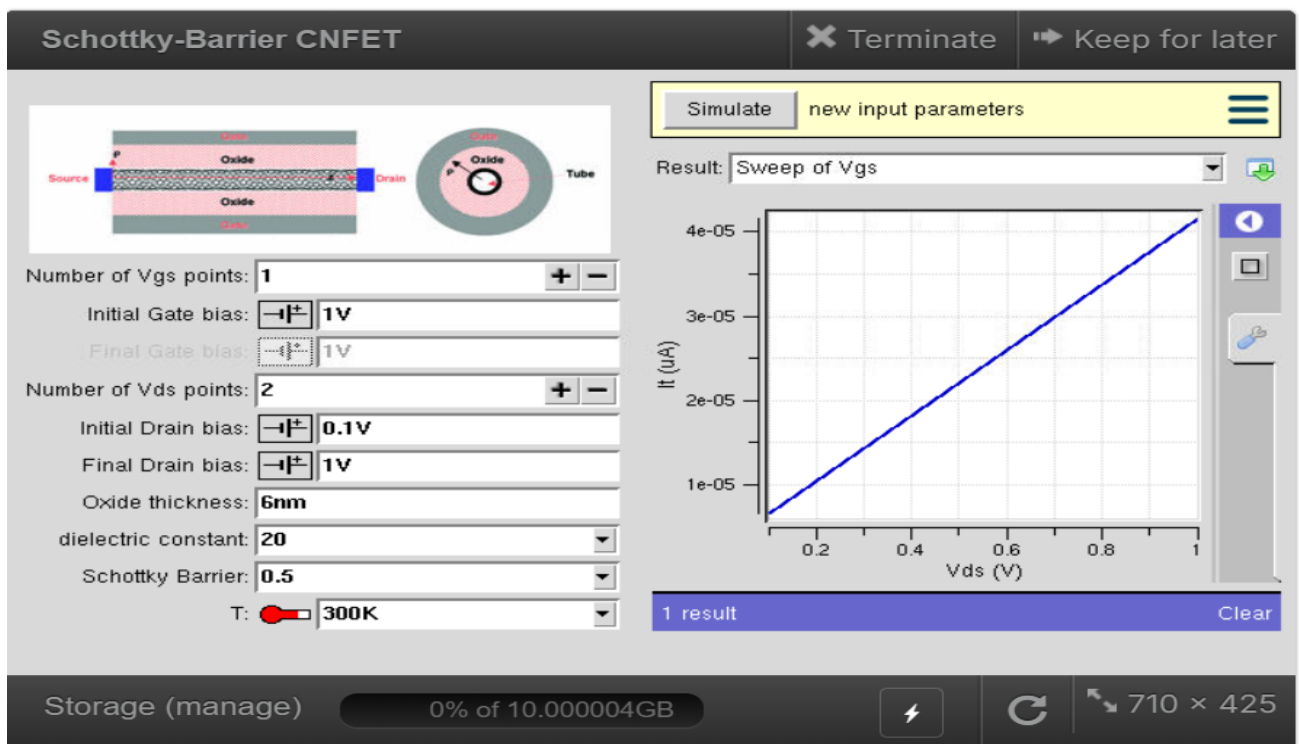
CNTFET



TEST CASE 1: thickness=4nm

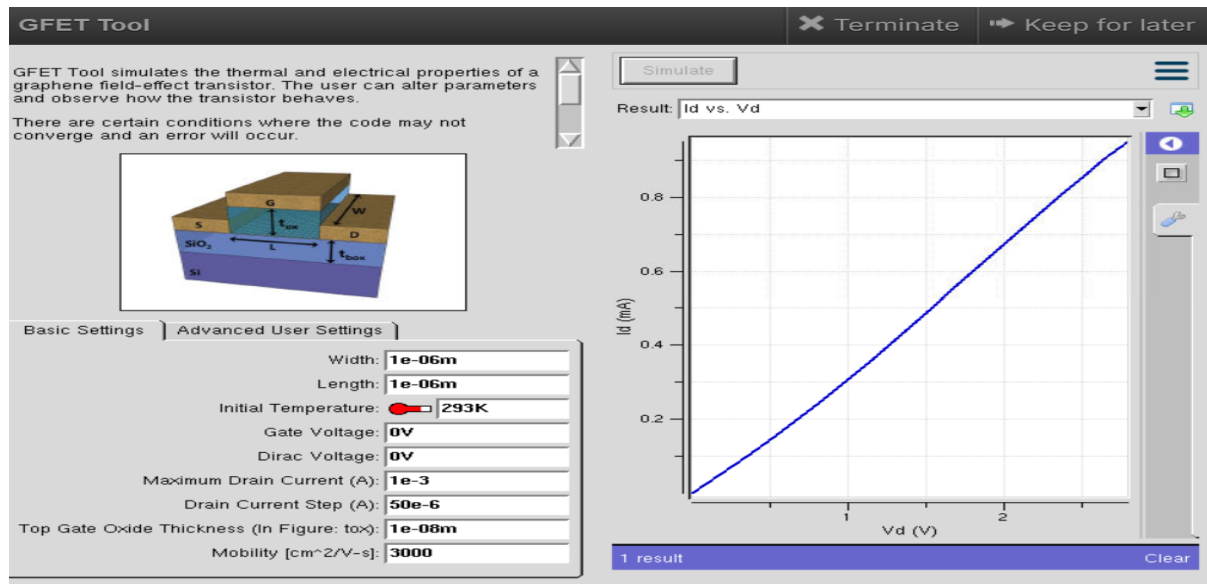


TEST CASE 2: Thickness=6nm

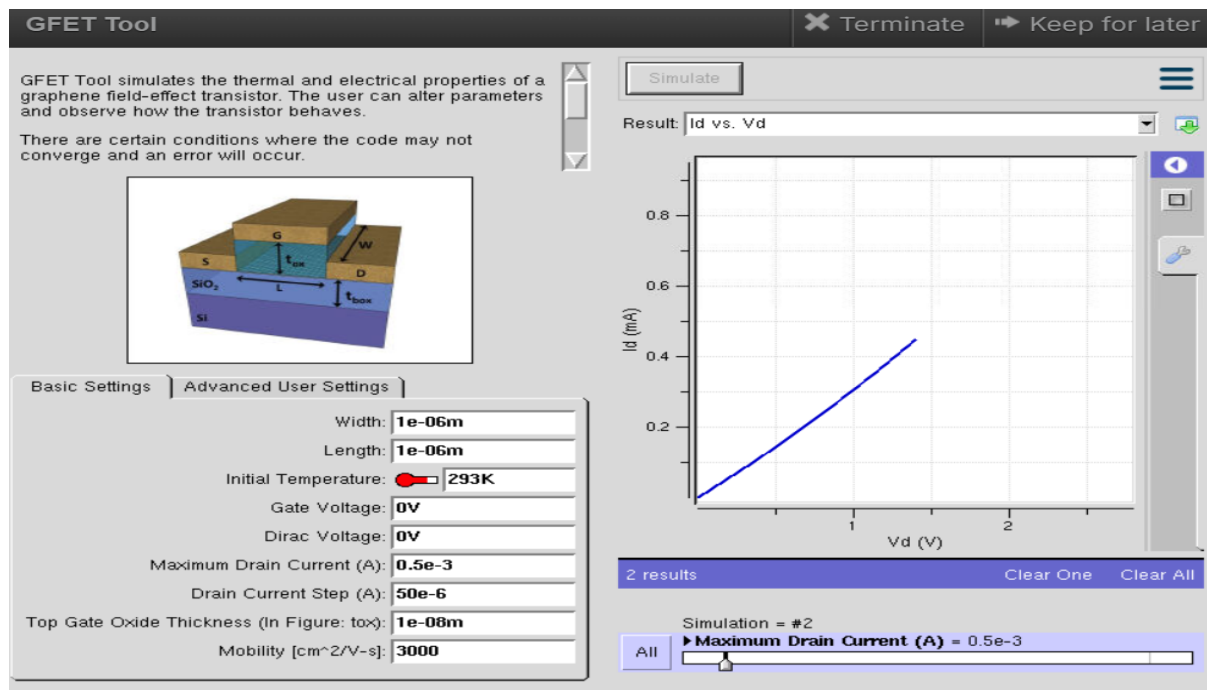


The graph shows that the drain current increases linearly with increasing drain voltage. This indicates efficient carrier transport through the carbon nanotube channel and stable operation of the Schottky-Barrier CNTFET. The high current confirms the excellent conductivity of carbon nanotubes.

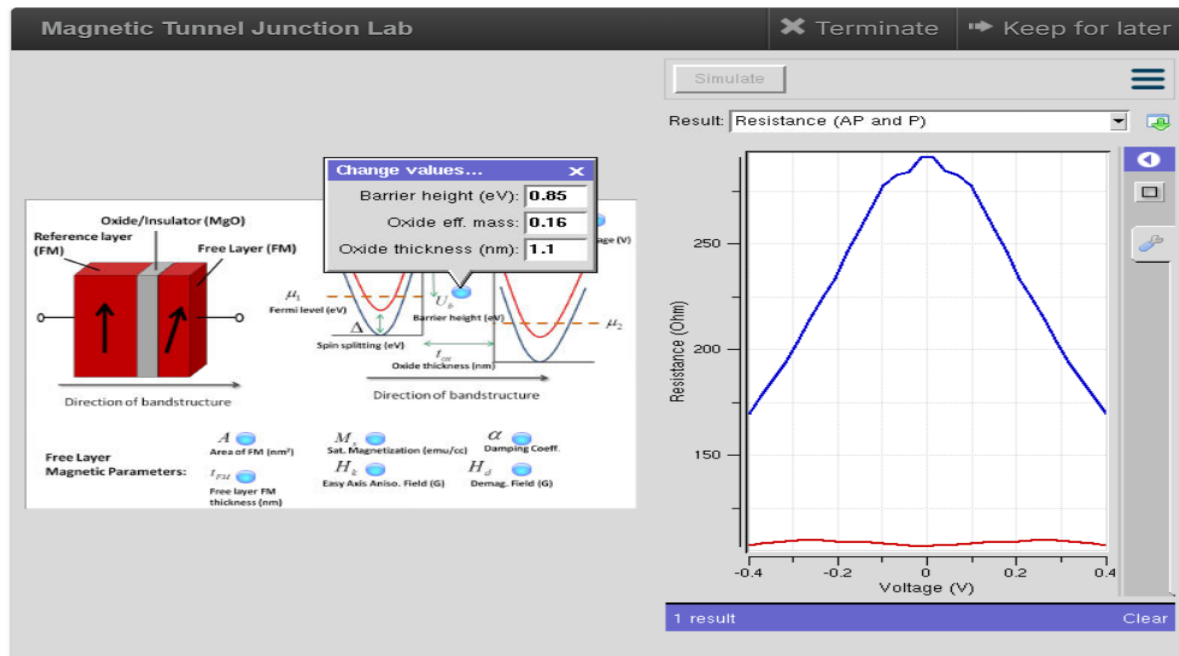
GFET : TEST CASE 1: MAXIMUM DRAIN CURRENT = 1e^{-3}



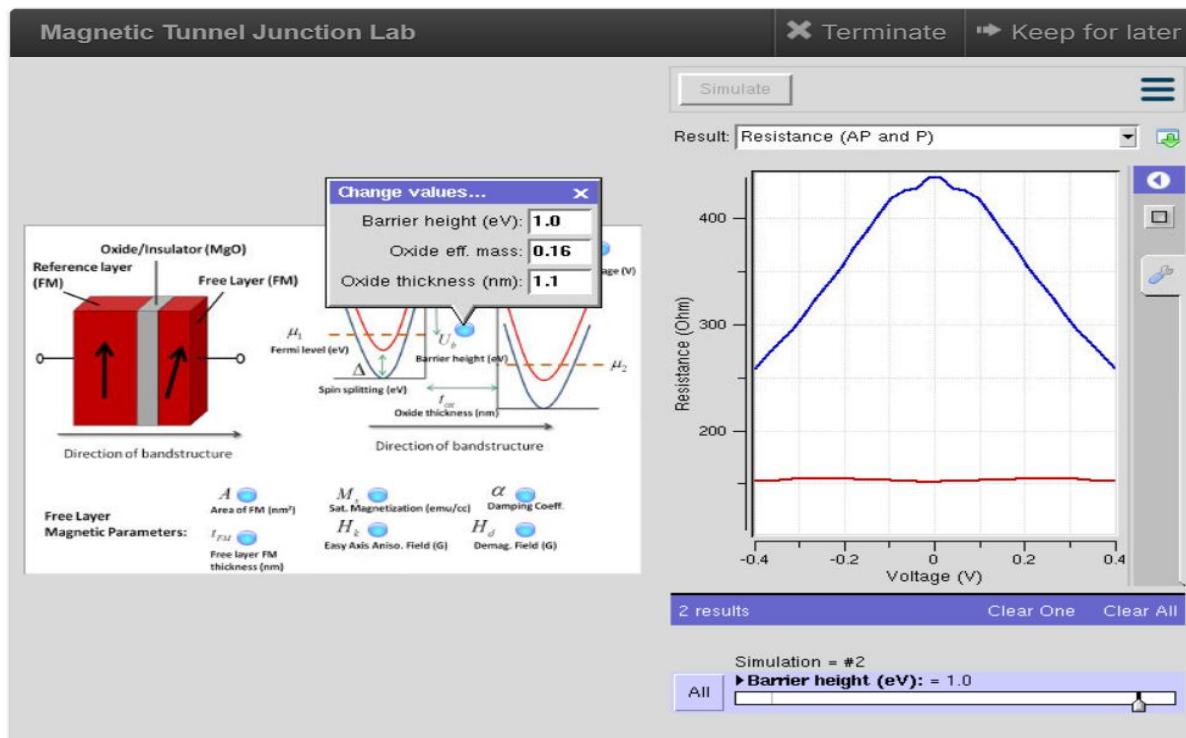
TEST CASE:2 MAXIMUM DRAIN CURRENT = 0.5e^{-3}



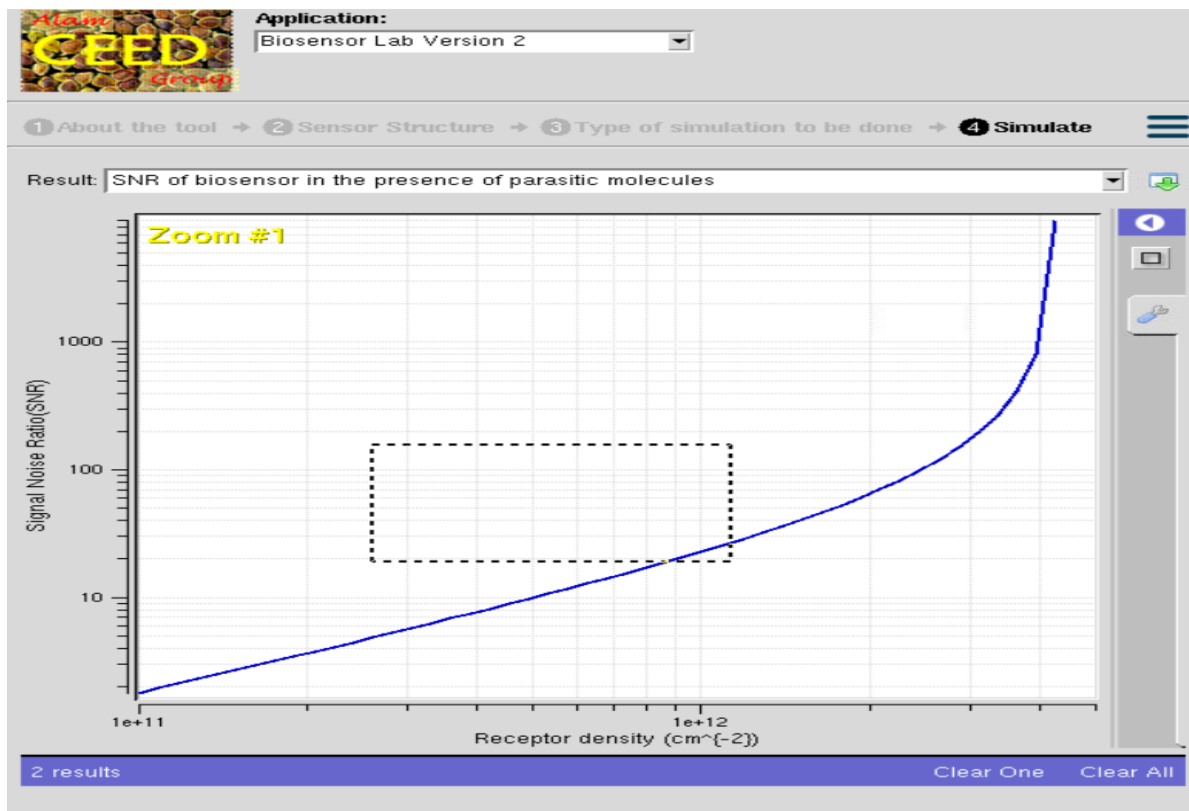
EXP 9: TEST CASE 1: Barrier Height=0.85eV



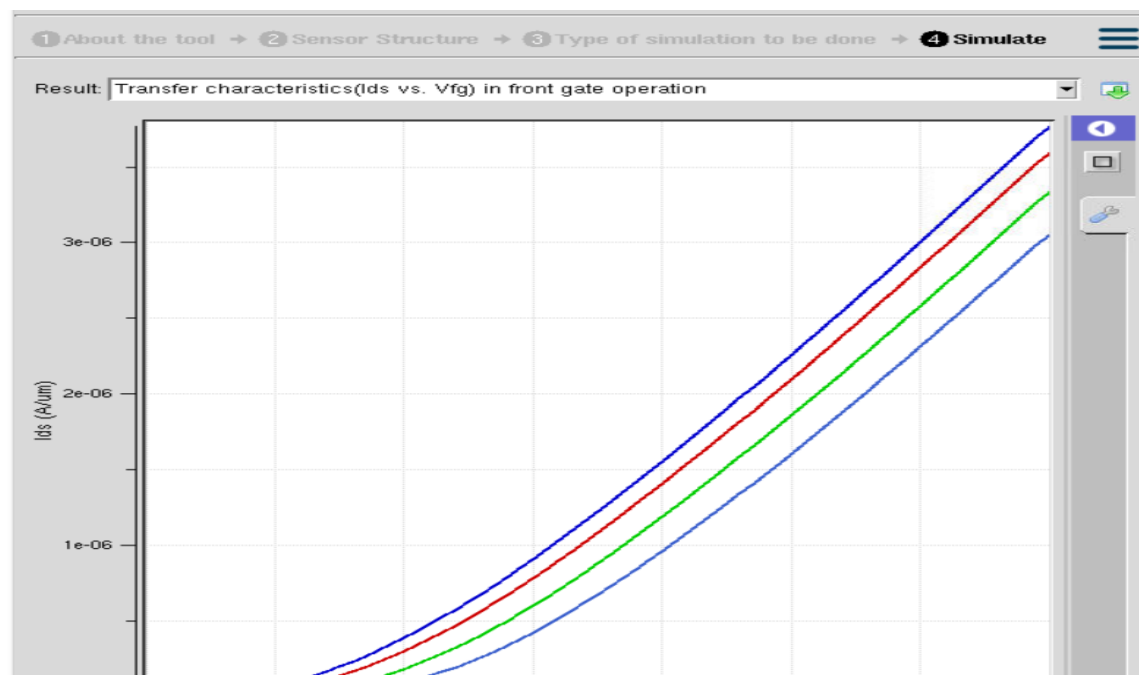
TEST CASE 2: Barrier Height = 1.0eV

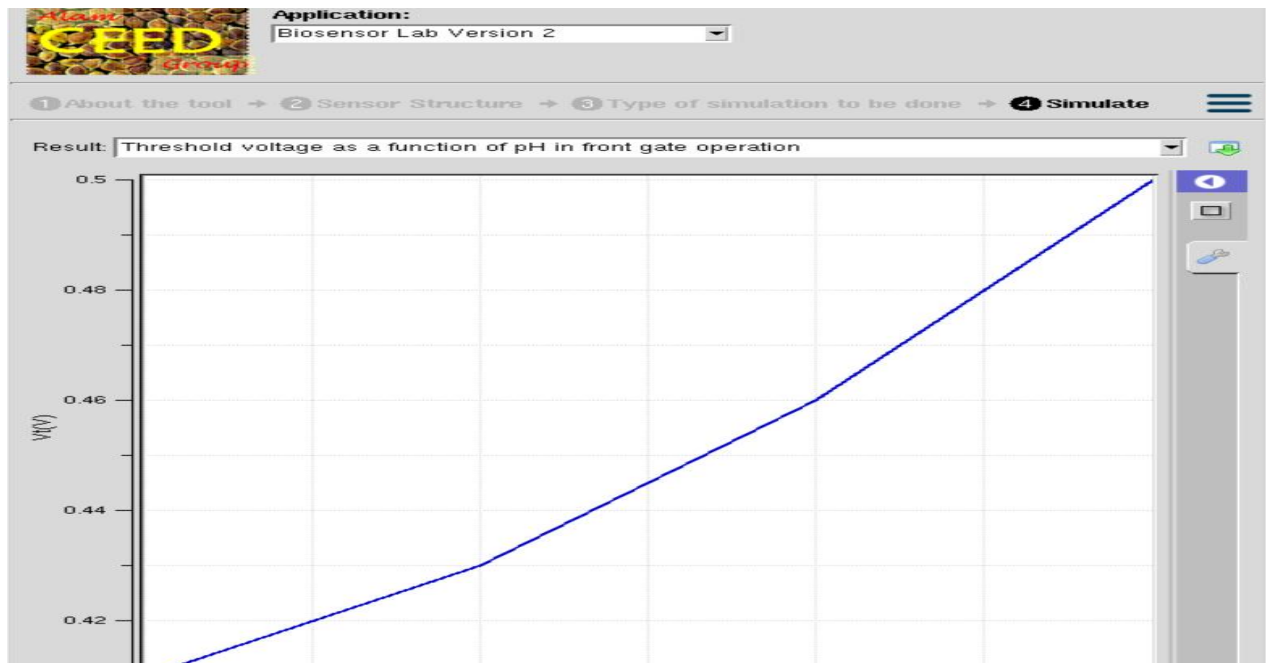


EXP 10:



TEST CASE 1:





The graph shows that the threshold voltage increases with increasing pH. This occurs because the surface potential at the gate changes due to hydrogen ion concentration. The linear variation indicates good pH sensitivity of the biosensor.